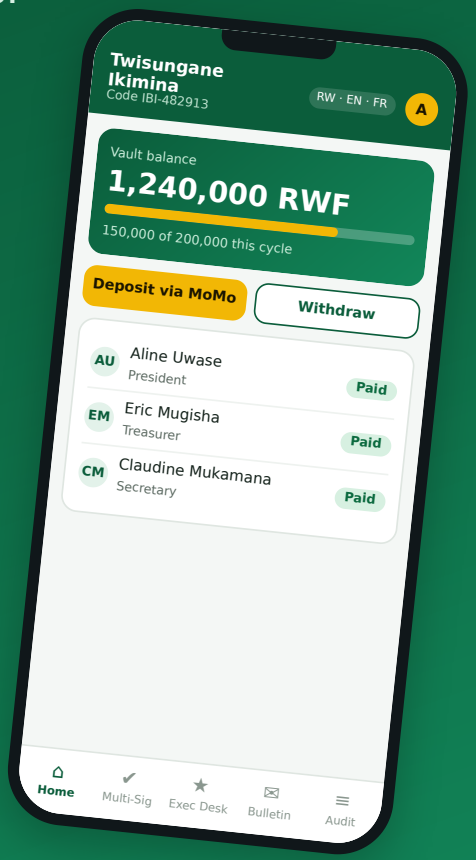
PITCH BOOK

Ibimina Community Banking

The trusted digital home for every Ikimina — shared savings, transparent decisions, and money that moves only when the group agrees.



Firebase Edition · Product, Technology & Roadmap
UI screens in this book are proposed design concepts

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B Demo script for presentations

HOW TO USE THIS BOOK

Each product chapter has two pages: a **phone page** and a **PC page**. Present the phone page first, because most Ikimina members will meet Ibimina on a phone. Gold numbered markers in each mockup match the numbered explanations beside it.

CHAPTER 1

The problem and the opportunity

Across Rwanda, neighbours, colleagues and families pool money in Ibimina (singular: Ikimina). It is one of the most trusted financial habits in the country — and one of the least supported by technology.

How an Ikimina works today

Members agree a regular contribution. Each cycle, the pooled money goes to one member in turn, or is kept as a shared fund from which members can borrow. The model works because people know each other. It strains when groups grow, when members live in different places, or when money has to be tracked precisely.

Paper and memory

Notebooks, cash boxes and WhatsApp messages hold the group's entire financial record. A lost notebook is a lost history.

One person holds the money

When a single treasurer controls the cash or the account, the group depends on one person's honesty, availability and memory.

Disputes about "who paid"

Without a shared, timestamped record, arguments over contributions, loans and interest are hard to settle fairly.

Distance excludes people

Members who move to another town or abroad often drop out because they cannot take part in meetings or see the books.

THE OPPORTUNITY

Mobile money is already part of daily life, and most members already carry a phone. What is missing is a purpose-built tool that respects how Ibimina actually work: shared decisions, rotating turns, small loans, and a visible record everyone can trust.

What a good solution must do

- Let a group start in minutes with only a phone number and a PIN.
- Make it impossible for one officer to move money alone.
- Show every member the same balance, the same votes and the same history, in real time.
- Speak the members' language: Kinyarwanda, English and French.

CHAPTER 2

The Ibimina solution

Ibimina Community Banking is a web app that gives every Ikimina a shared digital vault, a fair rotation, group-approved withdrawals and a permanent record, all synced live to every member’s phone.

Six things the app does

1 · Shared vault

One balance, visible to all, updated the moment money moves.

2 · Multi-sig withdrawals

A petition needs three-quarters of members to approve, including at least one ordinary member.

3 · Rotation schedule

Shows whose turn it is, and re-shuffles by lottery when the group wants.

4 · Emergency loans

Officers issue 30 or 60-day loans with 5% interest calculated automatically.

5 · Bulletin and chat

Announcements, structured meeting minutes and a live group conversation.

6 · Audit ledger

Every transaction gets a TXN ID and a timestamp, and is read-only.

Design principles

Principle	What it means in the product
Group first	Every screen answers “what does the group see?” before “what do I see?”.
Confirm before money moves	Every sensitive action asks for the 5-digit PIN again, even on an unlocked phone.
Simple enough for anyone	Five tabs, big buttons, plain words, and three languages.
Live, not eventually	A deposit or vote by one member appears on every other member’s screen within about a second.
Honest records	No editing or deleting of transactions; the ledger is the group’s memory.

IN ONE SENTENCE

Ibimina takes the trust that already exists in a savings group and gives it a transparent, tamper-resistant home.

CHAPTER 3

A month in the life of a group

The best way to understand Ibimina is to follow one group through a single cycle. The group below is fictional and is used throughout this book.

Twisungane Ikimina

8 members · 25,000 RWF monthly contribution · invite code IBI-482913

Cast

Aline (President), Eric (Treasurer), Claudine (Secretary), and five members including Diane, whose turn it is this month.

The cycle

1. **Start** — Aline creates the group and shares the invite code.
2. **Join** — Members join with the code, a phone and a PIN.
3. **Contribute** — Each member deposits 25,000 RWF via MoMo.
4. **Rotate** — The home screen shows Diane is due the payout.
5. **Decide** — A member files a petition; the group votes.
6. **Record** — Every movement lands in the audit ledger.

What changes for the group

Moment	Before Ibimina	With Ibimina
Paying in	Hand cash to the treasurer; hope it is written down.	Deposit via MoMo, confirm with PIN, see the vault grow instantly.
Asking for money	Private requests, side conversations, unclear approval.	A petition with amount, reason and recipient; a visible vote tally.
Meetings	Notes in a notebook, if anyone writes them.	Structured minutes with agenda, resolutions and actions, published to everyone.
Loans	Verbal agreements; interest argued later.	Principal, 5% interest and due amount calculated and recorded on issue.
Disagreements	Memory versus memory.	Timestamped ledger with TXN IDs.

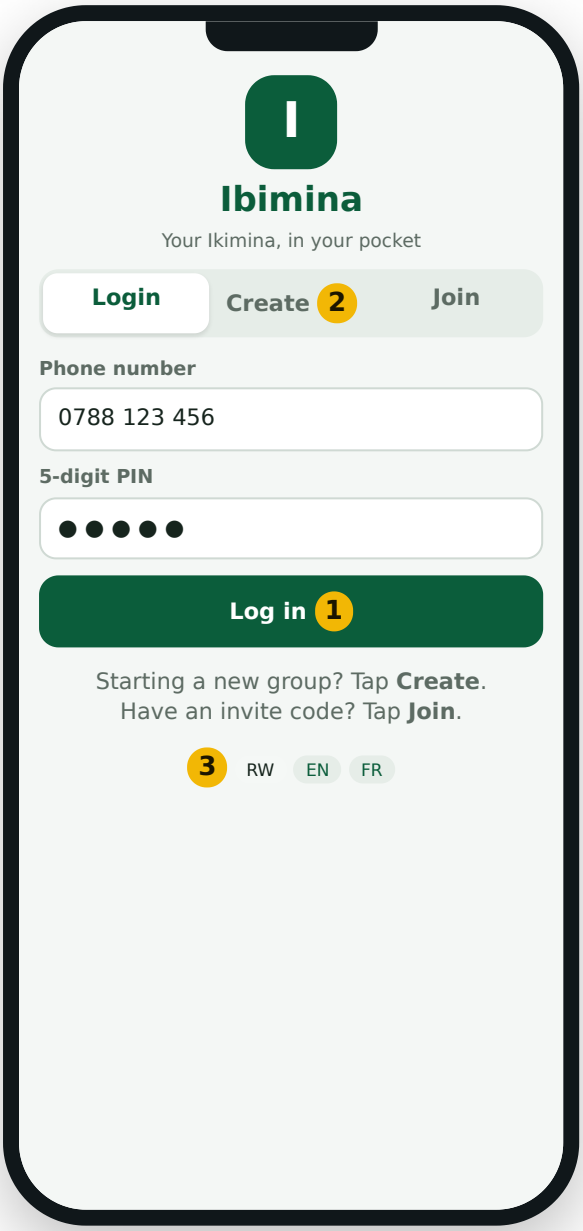
PITCH TIP

Tell this story out loud with a real group from your audience’s world. People remember a savings circle they recognise far better than a feature list.

CHAPTER 4

Welcome: login, create, join

The first screen asks only three things of a new member: a phone number, a PIN, and either a group name or an invite code. No passwords to forget, no email verification to wait for.



On a phone

What you are looking at

1. **Login** with phone number and a 5-digit PIN. The tabs above switch to **Create** (start a new group) or **Join** (enter an invite code like IBI-482913).
2. **Create** asks for group name, monthly contribution, your name and phone, and a PIN. You become the group’s **President** and receive an invite code to share.
3. **Language** can be switched between Kinyarwanda, English and French at any time; the choice is remembered on the device.

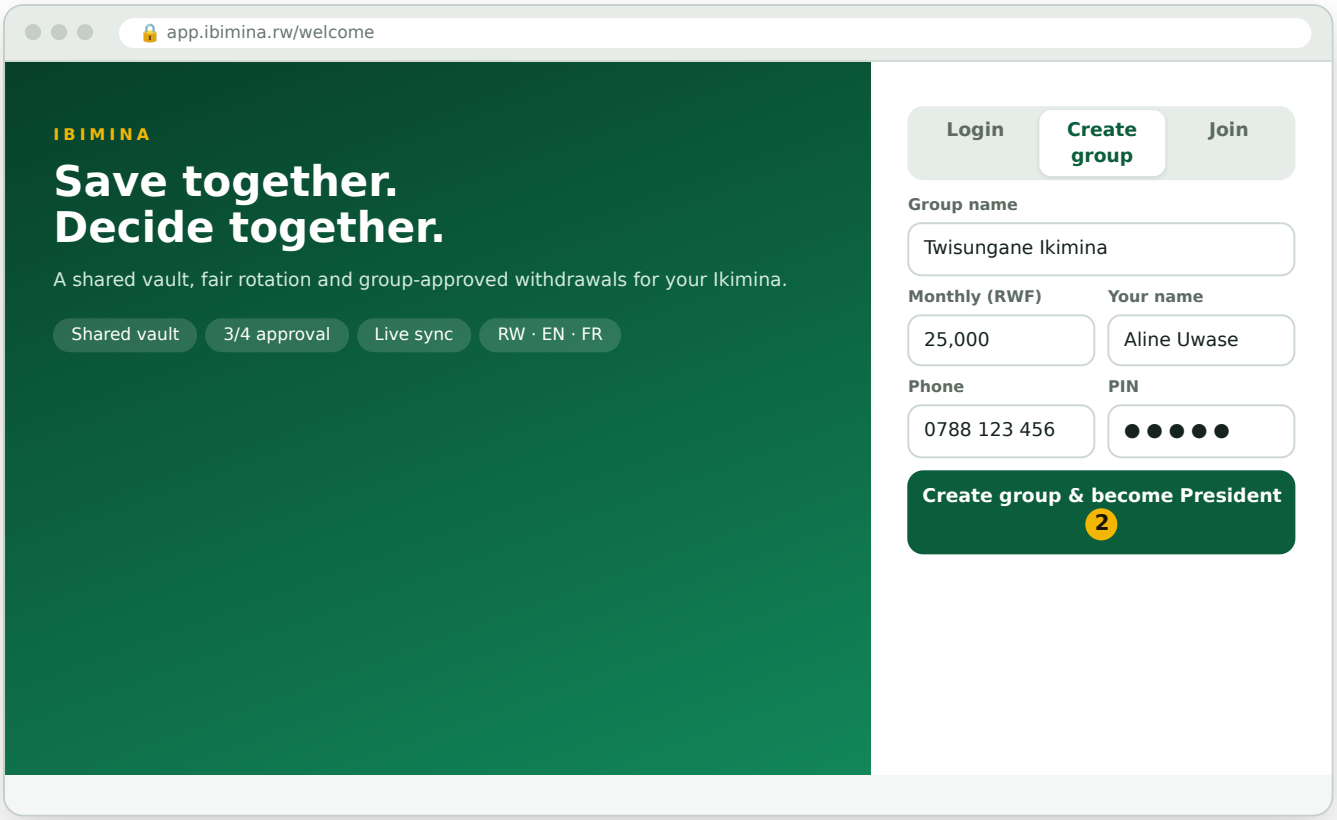
WHY THIS MATTERS

A phone number and PIN is how Rwandans already authorise mobile money. Members need no new habit, and no email account, to get started.

CHAPTER 4 · CONTINUED

Welcome: login, create, join · on a PC

On a larger screen the welcome page gains a brand panel. The same three tabs remain, so a President setting up the group at a desk follows the identical steps as a member on a phone.



On a PC or laptop

Behind the screen

- **Create** checks the phone format (078, 079, 072 or 073), then writes a new group and a President member to the database.
- **Join** looks up the invite code, assigns the next rotation position, and adds a Member.
- **Login** finds the member by phone and compares a SHA-256 hash of the PIN. The plain PIN is never stored.
- On success, live listeners attach to that group’s data.

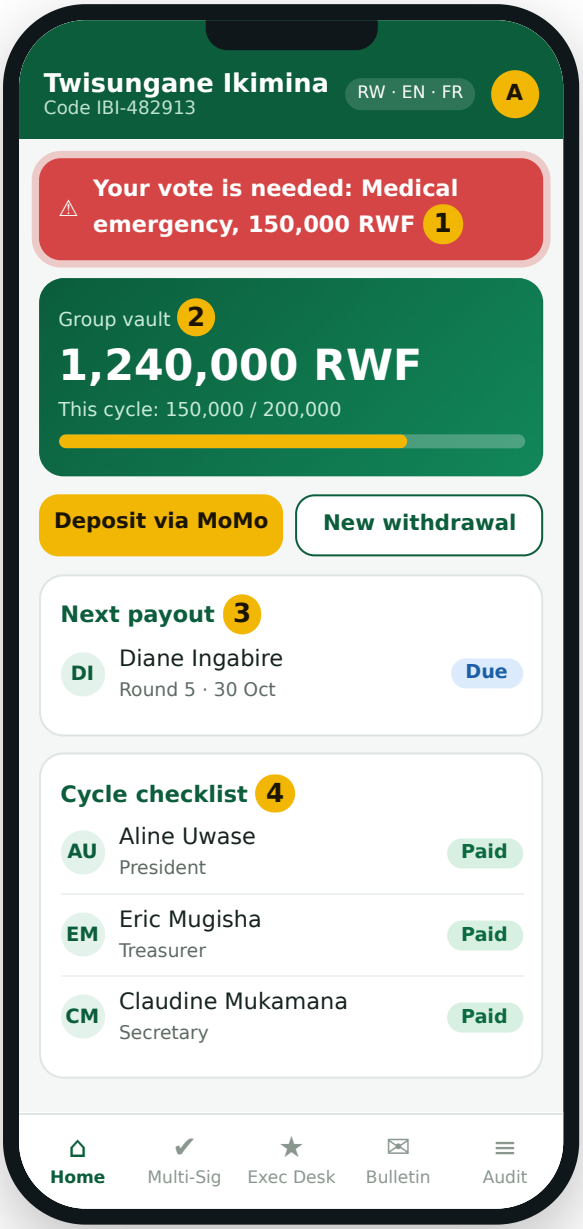
TALKING POINT

“Starting a group takes about a minute: name it, set the monthly amount, choose a PIN. Everyone else joins with a code.”

CHAPTER 5

Home dashboard

The Home tab answers the three questions every member asks first: how much do we have, whose turn is it, and has everyone paid?



On a phone

What you are looking at

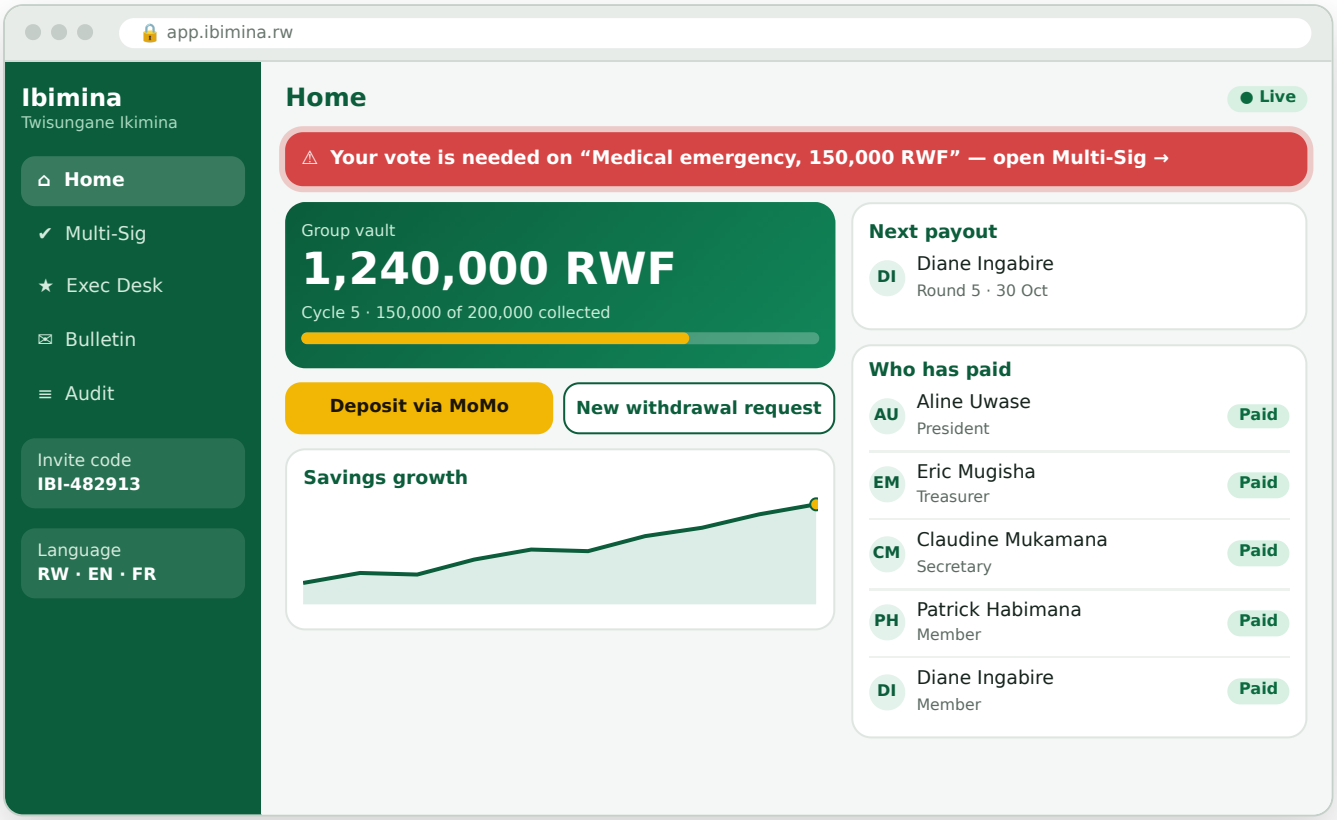
1. **Urgent banner.** If a withdrawal petition is waiting for your vote, a pulsing red banner takes you straight to it.
2. **Vault card.** The group balance, plus how much of this cycle’s target has been collected, shown as a progress bar.
3. **Rotation card.** Who receives the current payout and when.
4. **Cycle checklist.** A per-member Paid or Pending list, so nobody has to chase the treasurer to learn who is outstanding.

> **WHY THIS MATTERS**
>
> Transparency is the product. When every member sees the same vault and the same checklist, suspicion has less room to grow.

CHAPTER 5 · CONTINUED

Home dashboard · on a PC

On a computer the dashboard spreads across two columns and adds the savings-growth chart. The sidebar replaces the bottom tabs.



On a PC or laptop

Behind the screen

- The balance, rotation and checklist come from the group and member records in Firestore, kept live by real-time listeners.
- The savings chart is drawn with plain SVG, so there is no charting library to maintain.
- The urgent banner appears only when there is an active petition that *you* have not voted on.

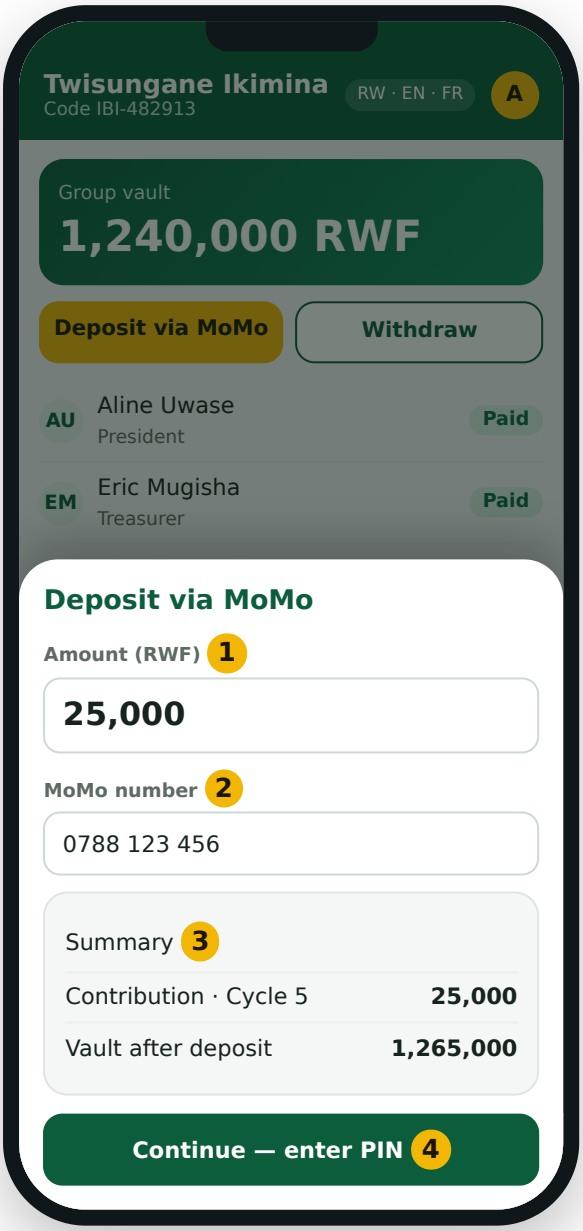
TALKING POINT

“Open the app and the first thing you see is the truth: the balance, the turn, and who has paid. No phone calls needed.”

CHAPTER 6

Depositing via MoMo

Paying in is two taps and a PIN. The member enters an amount, confirms the MoMo number, and authorises with the PIN. The vault, the cycle progress and the ledger update for everyone.



On a phone

What you are looking at

1. **Amount.** Pre-filled with the group’s monthly contribution, editable for part or extra payments.
2. **MoMo number.** The number the contribution comes from.
3. **Summary.** Shows exactly what the vault will hold after the deposit, so there are no surprises.
4. **PIN confirmation.** The single choke-point for every sensitive action; nothing is recorded until the PIN is correct.

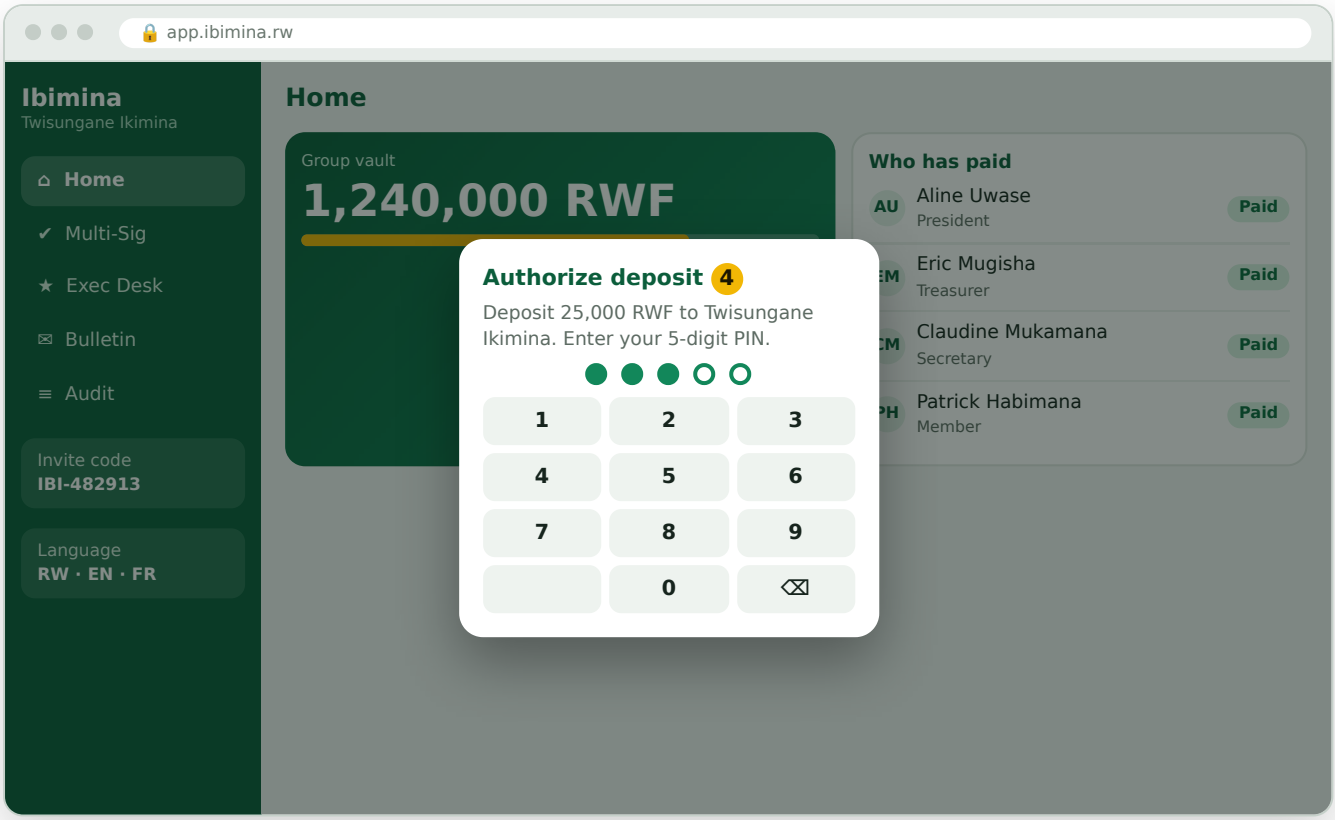
HONEST NOTE

In the current build, a deposit is **recorded** in the group’s books after PIN confirmation. A live payment request to the MoMo network is a planned integration (see Chapter 14), and should be described to partners that way.

CHAPTER 6 · CONTINUED

Depositing via MoMo · on a PC

On a computer the PIN prompt appears as a centred dialog above the dashboard, so the member never loses context.



On a PC or laptop

Behind the screen

- The PIN is hashed again and compared to the stored hash before anything is written.
- The vault balance, the member's total saved and their cycle status are updated.
- A ledger entry with a TXN ID is created.
- Every other member's screen refreshes through the live listeners.

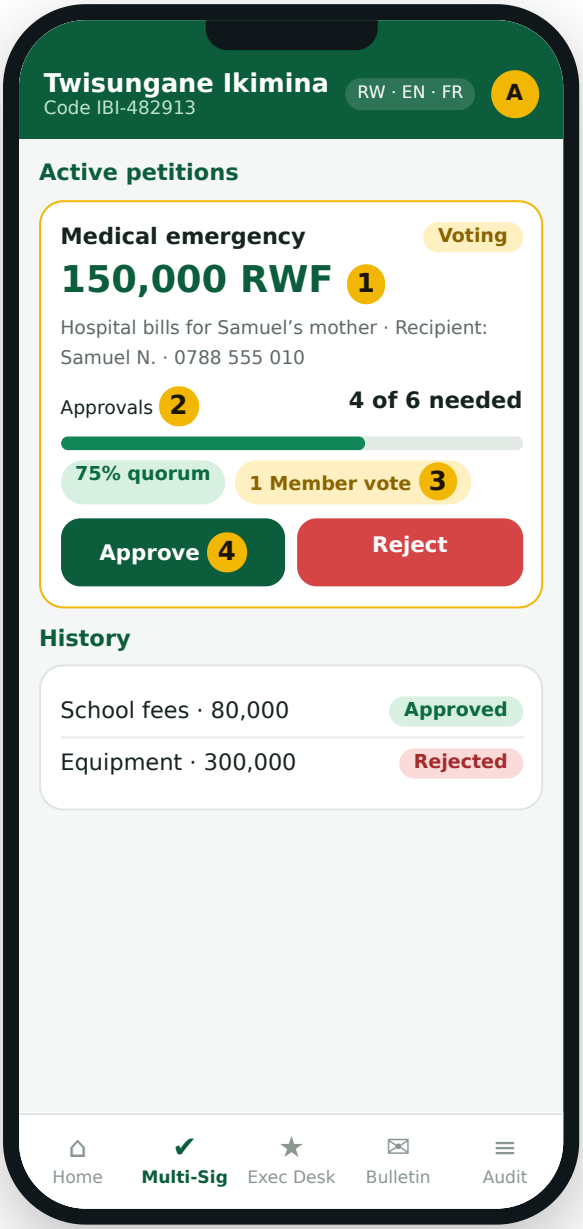
TALKING POINT

"Even if a phone is left unlocked on the table, nobody can move money without typing the PIN again."

CHAPTER 7

Multi-signature withdrawals

This is the heart of Ibimina’s trust model. Money leaves the vault only when the group agrees, and officers cannot approve alone.



On a phone

What you are looking at

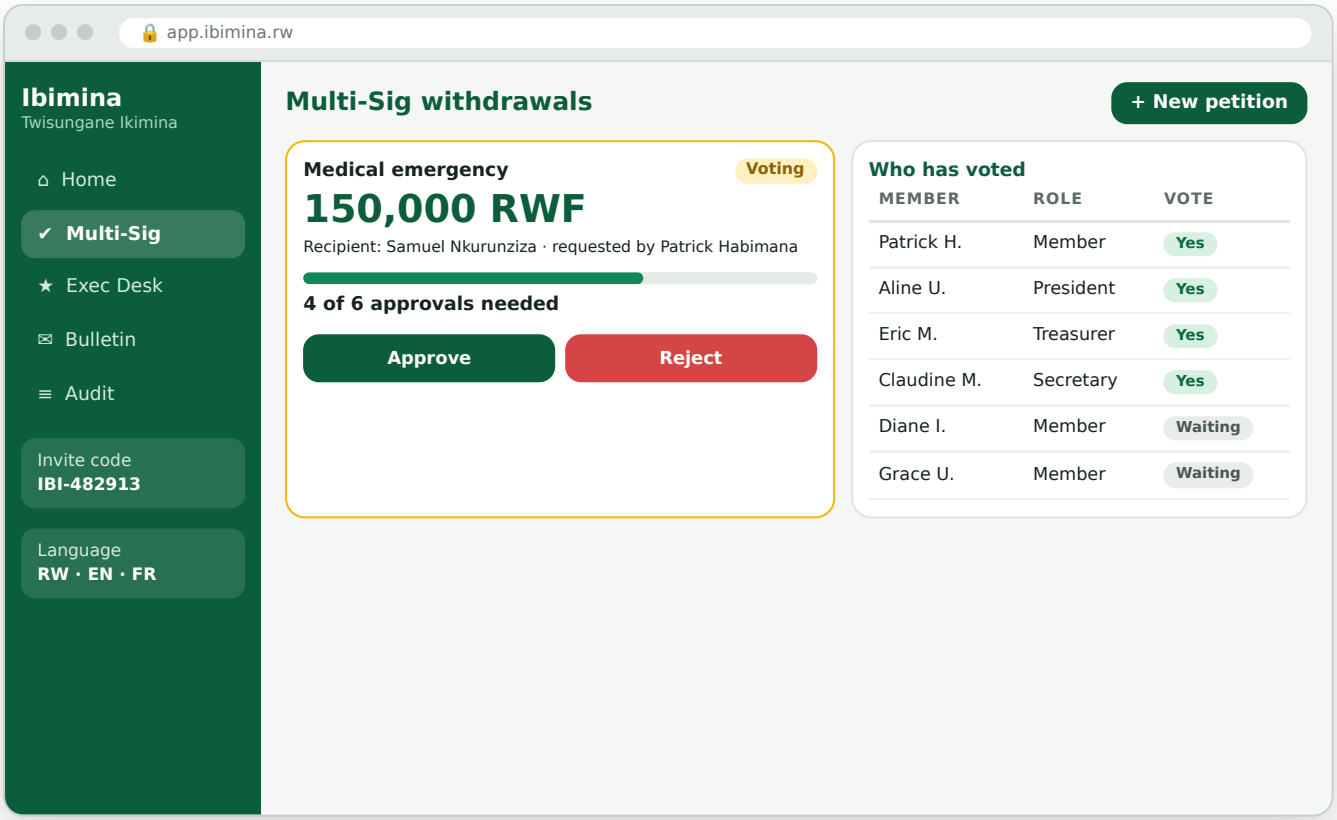
1. **The request.** Amount, reason and recipient are all visible to every member before anyone votes.
2. **Live tally.** A progress bar against the three-quarters rule. With eight members, six approvals are required.
3. **Member safeguard.** Once a group has four or more people, at least one approving vote must come from an ordinary Member, not just officers.
4. **Vote with PIN.** Approve or Reject opens the PIN prompt first. Votes cannot be cast by accident.

> **WHY THIS MATTERS**
> The fear in every shared fund is “what if the treasurer just takes it?” This screen is the answer. It makes collusion between officers alone structurally impossible.

CHAPTER 7 · CONTINUED

Multi-signature withdrawals · on a PC

On a computer the petition detail and the voter table sit side by side, which suits the President or Secretary running a meeting on a projector.



On a PC or laptop

Behind the screen

- A new petition starts with the creator’s own Yes vote already counted.
- Each vote rewrites the petition record, then the quorum rule is checked.
- When quorum is met, the vault is debited and a payout is written to the ledger.
- All members see the status change live.

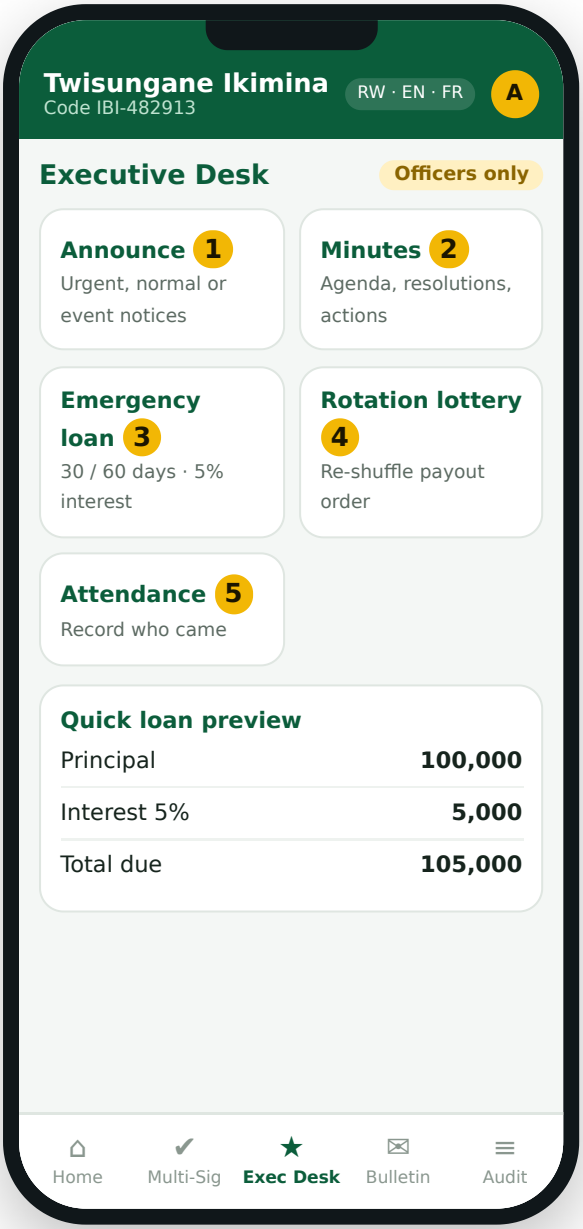
TALKING POINT

“Three out of four members must agree, and at least one must be an ordinary member. The officers cannot do it alone.”

CHAPTER 8

The Executive Desk

Officers get a separate workspace for the tasks that run a group: communicating, recording, lending and organising turns. Every tool asks for the PIN.



On a phone

What you are looking at

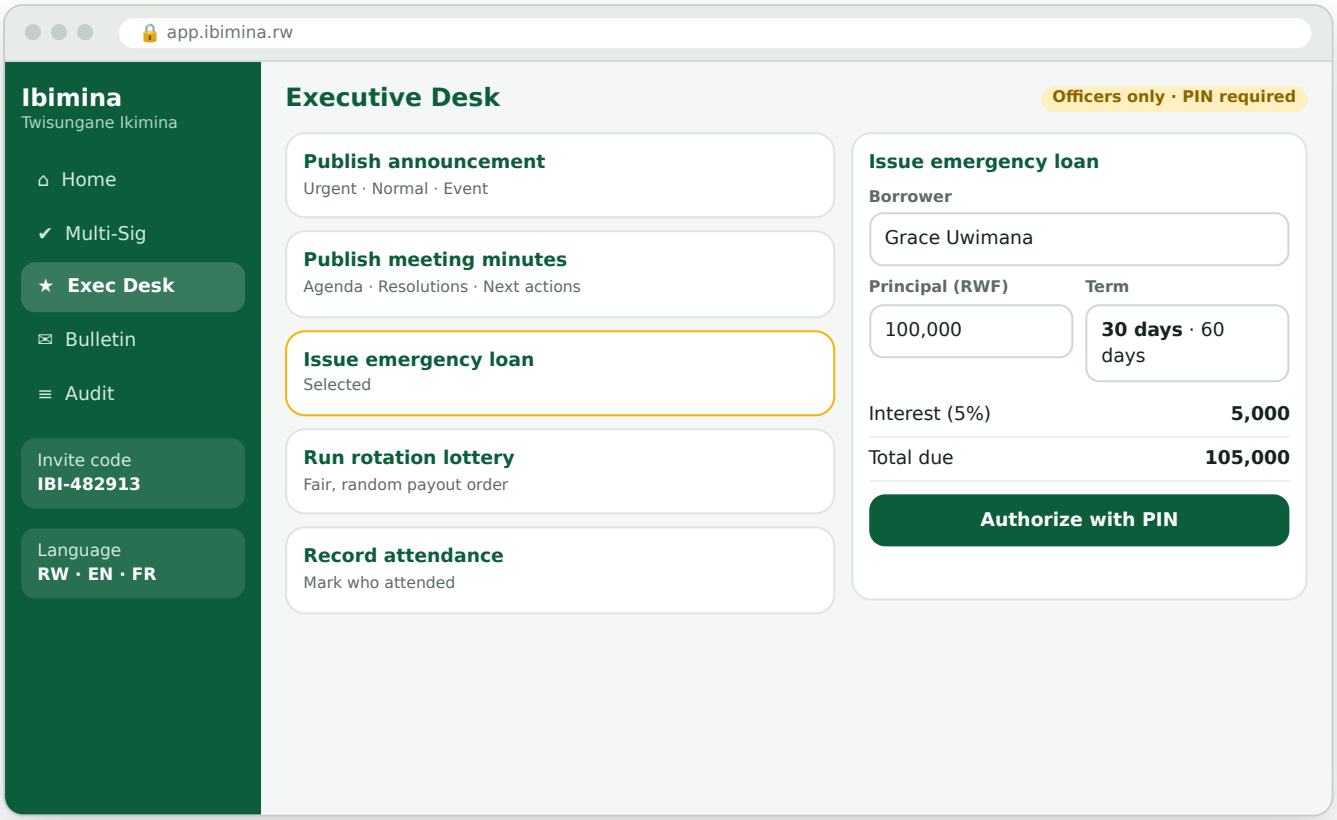
1. **Announce.** Post a notice with a priority: urgent, normal, event, or minutes.
2. **Minutes.** Structured meeting records so decisions are never lost.
3. **Emergency loan.** Choose borrower, amount and a 30 or 60 day term; interest of 5% and the total due are calculated automatically.
4. **Rotation lottery.** A fair re-shuffle of who receives the payout next.
5. **Attendance.** Record who attended a meeting.

> **WHY THIS MATTERS**
> Most Ikimina disputes are about information: what was decided, who was there, what was owed. This desk turns those moments into shared records.

CHAPTER 8 · CONTINUED

The Executive Desk · on a PC

On a computer the officer picks a tool on the left and fills in a roomy form on the right. This is the screen the President or Secretary uses during meetings.



On a PC or laptop

Behind the screen

- The rotation lottery updates every member's turn and the group's current recipient in **one atomic batch**, so nobody ever sees a half-shuffled order.
- Minutes and attendance are stored as special announcements, so they appear in the Bulletin feed automatically.
- Issuing a loan records principal, interest, total due, term and who authorised it.

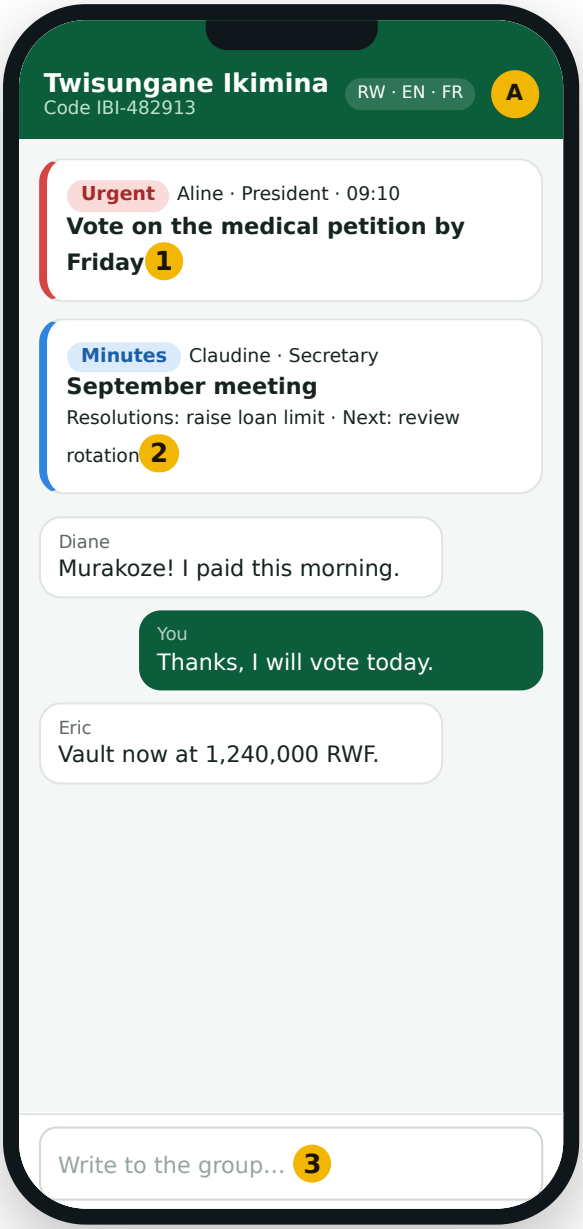
TALKING POINT

"Officers get tools, not power. The tools help them serve the group, and the group's vote still controls the money."

CHAPTER 9

Bulletin and group chat

Announcements and meeting records sit above a live group conversation, so the group’s official communication and everyday chat live in one place.



On a phone

What you are looking at

1. **Announcements.** Colour-coded by priority and labelled with author and role so members know who is speaking.
2. **Meeting minutes.** Structured records with agenda, resolutions and next actions.
3. **Group chat.** A simple live thread for the group. Messages are appended and are not edited or deleted.

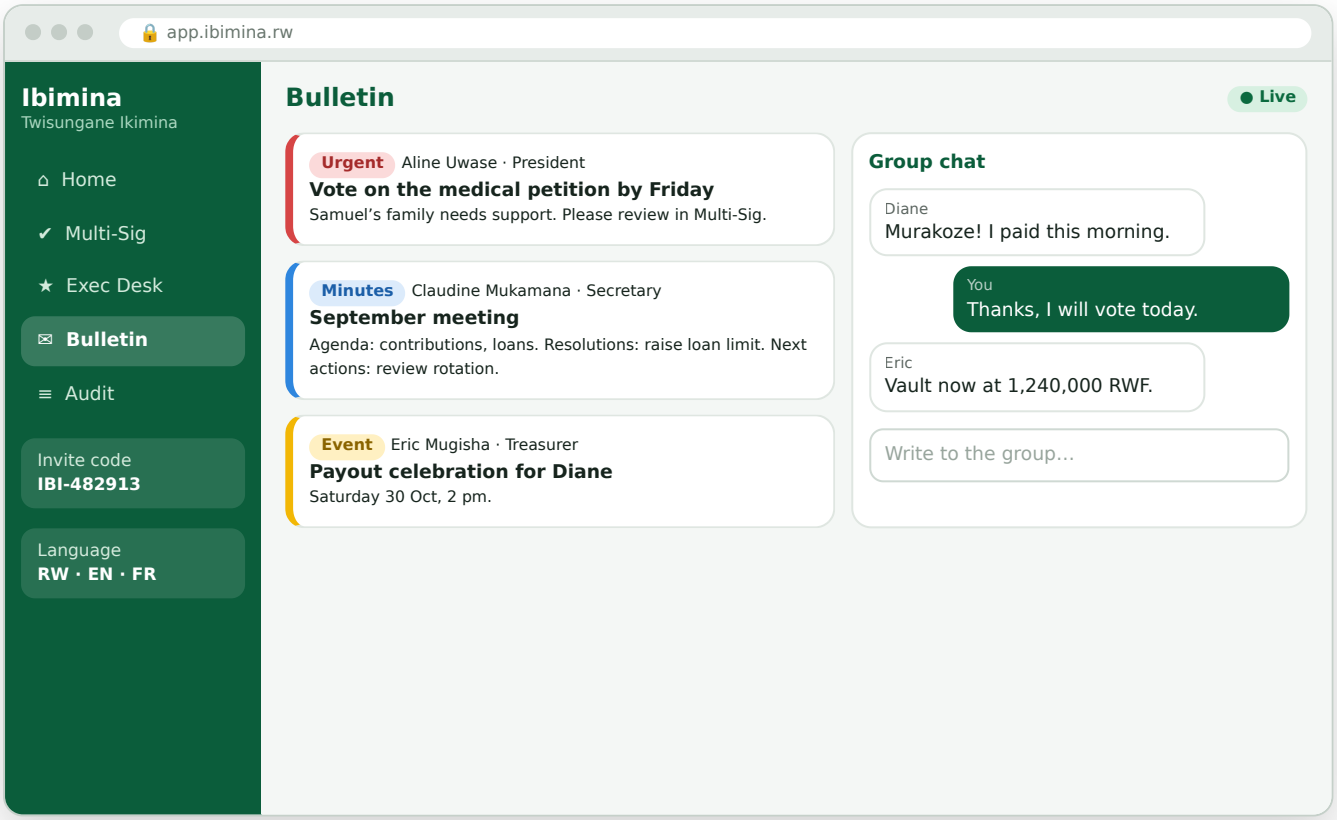
WHY THIS MATTERS

Groups already use WhatsApp for this, but it is separate from the money. Here the discussion and the decisions are in the same system as the vault.

CHAPTER 9 · CONTINUED

Bulletin and group chat · on a PC

On a computer the feed and the chat appear side by side, which makes meetings and long discussions comfortable.



On a PC or laptop

Behind the screen

- Announcements, minutes and attendance share one collection, so a single live feed covers all three.
- Chat messages record sender name and role.
- New items appear on every member's screen without refreshing.

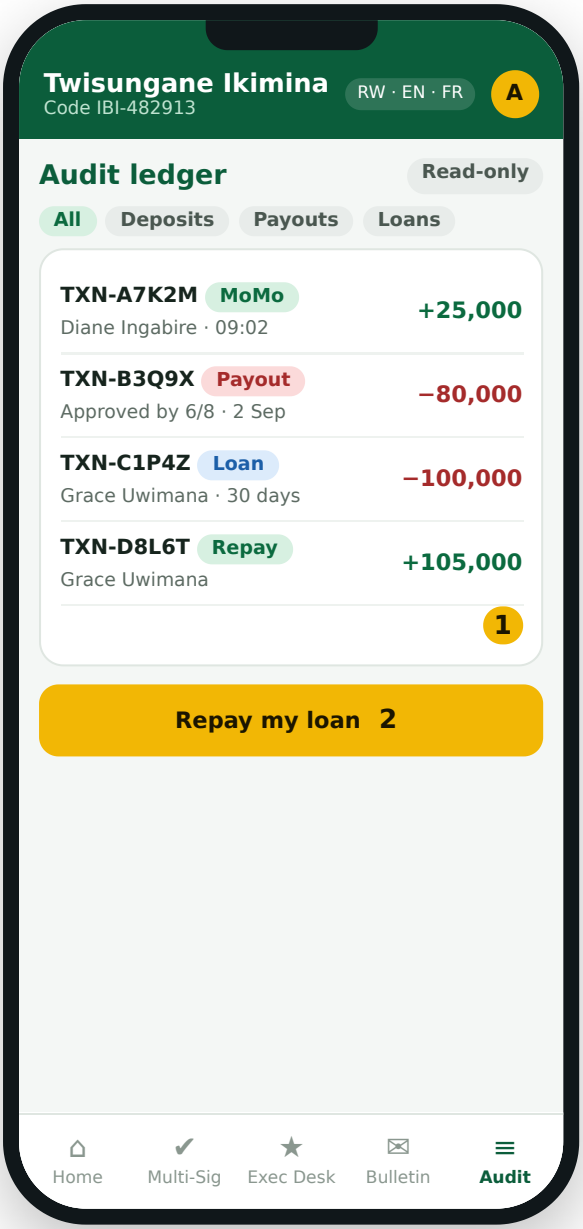
TALKING POINT

"Everything the group says and everything the group decides now lives next to the money it is about."

CHAPTER 10

Audit ledger and loans

The ledger is the group’s permanent memory: every movement of money, newest first, with a unique ID and a timestamp. Members can read it. Nobody can edit it.



On a phone

What you are looking at

1. **Ledger rows.** Each entry has a TXN ID, a type (MoMo, cash, payout, loan, repayment, penalty), who initiated it and the amount.
2. **Repay a loan.** A borrower repays from here, with PIN confirmation, and the vault is credited.

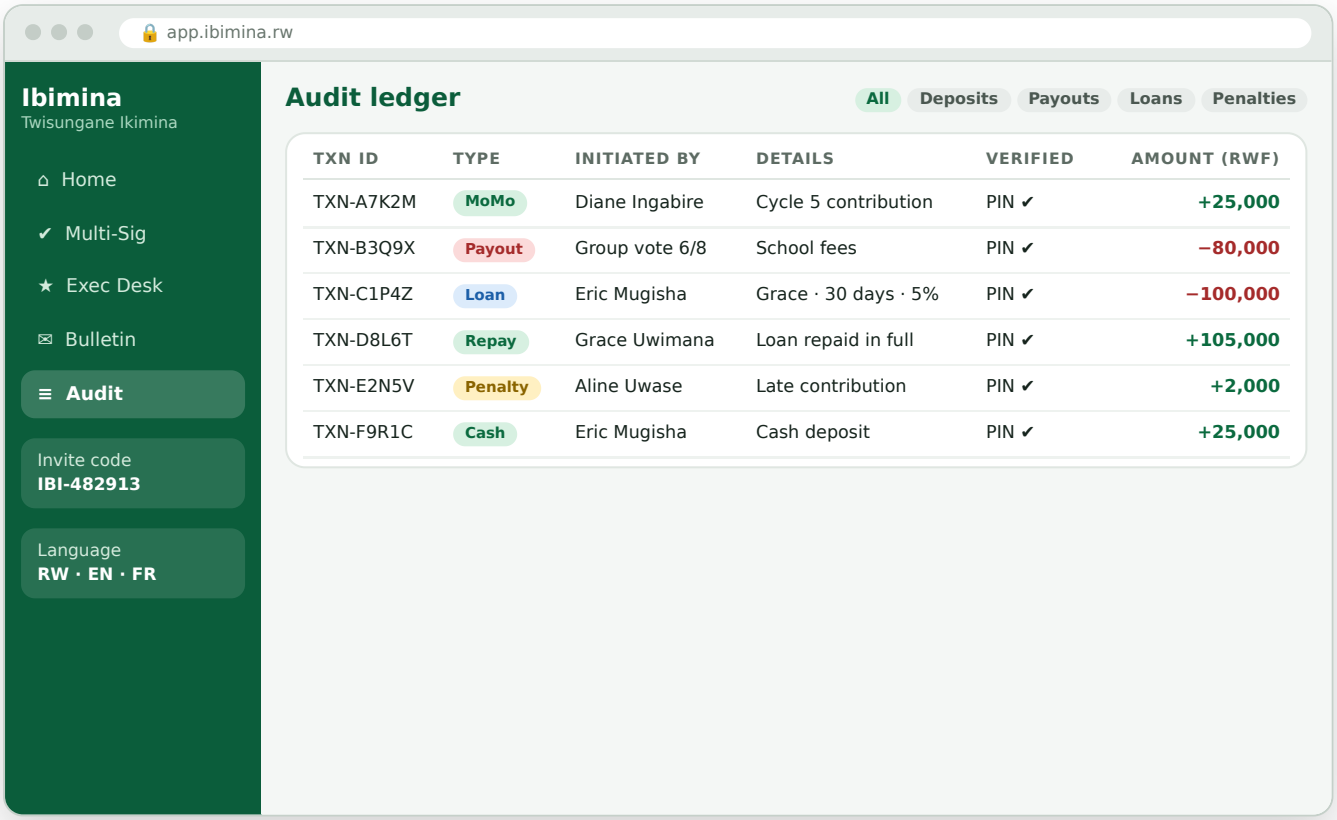
WHY THIS MATTERS

A permanent ledger turns “I think I paid” into “TXN-A7K2M, 09:02”. It ends the oldest argument in any savings group.

CHAPTER 10 · CONTINUED

Audit ledger · on a PC

On a computer the ledger becomes a full table, ideal for treasurers reviewing the month or for sharing screenshots at a meeting.



On a PC or laptop

Behind the screen

- The ledger is read-only for everyone.
- Entries are generated automatically when a deposit is made or a payout is approved.
- Loan status flips to **Repaid** when the borrower repays.

KNOWN GAP, PLANNED FIX

Loan issuance and repayment currently change the vault but do not yet write ledger lines. Closing this hole is the first item in the roadmap (Chapter 14).

TALKING POINT

“If it moved money, it is on the ledger, with a time and a name. That is the promise.”

CHAPTER 11

Trust and security model

Ibimina’s promise is trust. The design uses three layers: who you are, what you confirm, and what the group allows.

1
Identity
Phone number plus a 5-digit PIN. The PIN is stored only as a SHA-256 hash.

2
Confirmation
Every sensitive action re-checks the PIN at the moment of action: voting, depositing, petitions, loans, announcements, repayments.

3
Group consent
Withdrawals need three-quarters approval and at least one ordinary Member’s vote.

The withdrawal path

1. **Petition** — A member states amount, reason and recipient.
2. **Vote** — Each vote needs the voter’s PIN.
3. **Quorum** — 75% and one Member vote reached.
4. **Payout** — Vault debited, ledger written.

What each layer protects against

Risk	Protection
An unlocked phone on a table	PIN re-entry for every sensitive action.
A rogue treasurer	Officers cannot reach quorum alone.
“I never agreed to that”	Voter names, roles and decisions are visible on the petition.
Disputed history	Read-only, timestamped ledger.
Someone reading the database for PINs	Only hashed PINs are stored.

BE CANDID WITH PARTNERS

These protections live in the app today. The next step is to enforce the same rules **on the server** with Firestore security rules, so they hold even against someone bypassing the app. Chapter 14 explains the plan.

CHAPTER 12

Architecture and technology

Ibimina is a single-page web app that talks directly to Google’s Firestore. There is no custom server to run, which keeps hosting simple and costs low for an early product.

What members see
React 18 · TypeScript · Tailwind CSS · lucide icons

Shared state
React Context holds group, members, petitions, announcements, chat, loans, ledger

Live connection
Seven real-time listeners per logged-in member

Database
Google Firestore (Firebase SDK v11), accessed directly from the browser

Technology at a glance

Layer	Choice
Framework	React 18, function components and hooks
Language and build	TypeScript, Vite 6
Styling	Tailwind CSS 3
Database	Cloud Firestore
Identity	Phone number + hashed PIN (custom, not Firebase Authentication)
State	React Context
Languages	Kinyarwanda, English, French; the choice is saved on the device

STRENGTHS
Fast to build, cheap to host, instant live updates, works in any modern phone browser.

TRADE-OFFS
No server-side rules yet, so security checks must move into Firestore rules and, later, trusted server code.

CHAPTER 13

The data model

Seven flat collections, each document tagged with its group. It is simple to query and easy to extend.

Collection	Holds	Used by
`groups`	Name, invite code, vault balance, cycle round, target payout, monthly contribution, current recipient, next payout date	Home, every money action
`members`	Name, phone, role, hashed PIN, cycle status, total saved, rotation order	Login, checklist, rotation
`petitions`	Title, reason, amount, recipient, status, votes, required ratio	Multi-Sig
`announcements`	Title, content, author, priority, timestamp; also minutes and attendance	Bulletin, Executive Desk
`chatMessages`	Sender, role, message, timestamp	Bulletin
`loans`	Borrower, principal, interest, total due, term, status, authoriser	Executive Desk, Audit
`auditLogs`	TXN ID, type, amount, initiator, details, timestamp, PIN verified	Audit ledger

How data reaches the screen

When a member logs in, the app opens seven live listeners, one per collection, each filtered to that member’s group. When anyone in the group writes, every listener fires and every screen updates, with no refresh button.

Reads

Screens are fed entirely by live listeners. Only login and invite-code lookups use one-off reads.

Writes

New documents, partial updates, and one atomic batch for the rotation lottery.

WHY FLAT COLLECTIONS?

Phone numbers must be unique across all groups, which needs a database-wide search. Flat collections with a `groupId` make that straightforward.

CHAPTER 14

Production readiness and roadmap

Ibimina is production-shaped, not yet production-ready. This chapter lists exactly what stands between today's build and a group trusting it with real savings, and how each item will be closed.

#	Gap today	Planned fix	Phase
1	No Firestore security rules shipped; client-side PIN checks can be bypassed	Write and test security rules; enforce group membership and quorum on the server.	**Now**
2	No PIN attempt limiting (100,000 combinations)	Lockout and delay after repeated failures; move verification server-side.	**Now**
3	Deposit and payout update the balance by read-then-write	Use Firestore transactions so simultaneous deposits cannot overwrite each other.	**Now**
4	Loans and repayments are missing from the ledger	Write a ledger entry for every balance change.	**Now**
5	Refresh logs you out	Persist a secure session so the app can be reopened quickly.	Next
6	One phone number can belong to one group only	Allow membership in several groups per person.	Next
7	Email collected but unused	Use it for recovery and receipts, or remove the field.	Next
8	Login screen styling differs from the rest of the app	Unify the visual identity, with light and dark themes.	Next
9	MoMo deposits are recorded, not requested from the network	Integrate a MoMo payment provider for real collection and payout.	Later

WHY SHARE THIS?

Investors and partners respect a team that knows its risks. A short, specific list shows engineering discipline and gives them a clear picture of what funding or partnership will deliver.

1. **Now** — Make money-handling safe.
2. **Next** — Make daily use smooth.
3. **Later** — Connect to the real money rails.

CHAPTER 15

Sustainability and impact

Ideas for how Ibimina can pay for itself and what success should look like. These are proposals for you to refine; no figures below are claims about existing users or revenue.

Possible revenue models

Free core, paid extras

Basic group management is free. Advanced reports, exports and multiple groups sit in a paid plan.

Small transaction fee

A small, transparent fee on payments once real MoMo integration exists.

Partner bundles

SACCOs, NGOs and savings programmes pay to onboard and support their member groups.

Group financial profile

With consent, a group’s transparent ledger could support access to micro-credit.

Impact measures to track

Measure	Why it matters
Groups created and still active after 3 months	Shows whether the product fits real routines.
Share of contributions recorded on time	Shows improved discipline and transparency.
Withdrawals approved through quorum	Shows the trust model is actually used.
Disputes resolved by the ledger	Tells the human story to funders.
Members reached outside the original town	Shows distance no longer excludes people.

WHO BENEFITS

Members gain safety and visibility. Officers gain credibility and relief from record-keeping. Communities gain a way to build collective savings that outlasts any one treasurer.

FILL IN BEFORE PRESENTING

Add your own pilot numbers here when you have them: groups, members, total savings tracked. Real numbers are always stronger than projections.

CHAPTER 16

The ask and next steps

Every pitch ends with a clear request. Choose the one that fits your audience, and delete the rest.

For funders

Support to complete the “Now” security phase, run pilots with real groups, and integrate MoMo.

For community partners

Introductions to 5–10 Ibimina willing to pilot the app and give honest feedback.

For technical partners

Review of the security rules and architecture; MoMo integration expertise.

For mentors

Guidance on regulation, SACCO relationships and sustainable pricing.

A 90-day plan

When	Milestone
Days 1–30	Security rules, PIN lockout, transactions and complete ledger. Internal testing with a mock group.
Days 31–60	Session persistence, multi-group membership, unified design. Pilot with 2–3 real Ibimina.
Days 61–90	Feedback round, MoMo integration design, impact metrics reported.

CLOSING LINE

“Ibimina does not replace the trust between neighbours. It protects it, and lets it travel.”

Contact

Add your name, role, phone, email and the project link here before printing or sharing this book.

APPENDIX A

Glossary

Term	Meaning
Ikimina / Ibimina	A savings group in Rwanda / the plural form.
Vault	The group’s shared balance.
Cycle	One round of contributions ending in a payout to one member.
Rotation	The order in which members receive the payout.
Petition	A request to withdraw money, needing group approval.
Multi-sig	Approval by several people instead of one. Ibimina requires 75%, including one ordinary Member.
Quorum	The minimum approval required for a petition to pass.
MoMo	Mobile money.
PIN	The member’s 5-digit code used at login and to confirm actions.
Hash	A one-way scrambled form of the PIN, so the real PIN is never stored.
TXN ID	The unique reference for every ledger entry.
Firestore	Google’s cloud database that keeps every member’s screen in sync.
Security rules	Server-side rules that decide who may read or write which data.

APPENDIX B

Demo script for presentations

A five-minute live walkthrough. Use two phones, or one phone and a laptop, so the audience sees changes appear live on the second device.

Min	Show	Say
0:00	Create a group (Chapter 4)	“A President starts a group with a phone number and PIN.”
0:45	Join from the second device with the code	“Members join with a code. No email, no passwords.”
1:30	Deposit via MoMo with PIN (Chapter 6)	“Watch the other screen update instantly.”
2:30	File a petition and vote (Chapter 7)	“Money leaves only when three-quarters agree, including a regular member.”
3:30	Issue an emergency loan (Chapter 8)	“Interest and total due are automatic.”
4:15	Open the ledger (Chapter 10)	“Every movement, a time and a name.”
4:45	Roadmap and ask (Chapters 14 and 16)	“Here is what we will finish and what we need.”

BEFORE YOU PRESENT

Create a demo group with fictional names, pre-load a few transactions, and test the Firebase connection on the venue’s network. Have screenshots of Chapters 5 to 10 ready in case the internet fails.